

Executive function and maladaptive coping in a young adult cohort of individuals with congenital heart disease

Sarah A. Mathena^{a,*}, Vanessa F. Eller^{b,1}, Scott E. Klewer^{a,1}, Jennifer G. Andrews^{a,1}

^a Department of Pediatrics, University of Arizona, Tucson, AZ, USA

^b University of Arizona College of Medicine – Phoenix Internal Medicine Residency Program, USA

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ABSTRACT

Background: Young adults with congenital heart disease (CHD) are more likely to exhibit executive dysfunction and maladaptive coping yet are less likely to get diagnosed than the general population for psychiatric disorders. The etiology of psychiatric diagnosis is inconclusive amongst CHD populations, and more research is needed to evaluate risk factors contributing to psychiatric disorders, including executive function, coping mechanisms, anxiety, depression, fatigue and sleep disorders, parental style/involvement, and sense of coherence.

Objectives: The study proposed to demonstrate the presence of risk for psychiatric diagnoses amongst young adults with CHD and evaluate factors contributing to mental health and executive dysfunction comorbidities.

Methods: We used a mixed methods study of quantitative scales and a guided interview on a cohort of 11 young adults (18–24) with CHD who are considered in the American Heart Association high-risk group for developmental disorders or disabilities. We assessed for risk of anxiety and depression, maladaptive and disengagement coping, executive dysfunction, and other factors associated with mental health using the Responses to Stress (RSQ), Patient Reported Outcomes Measurement Information System (PROMIS), and the Amsterdam Executive Function Inventory (AEFI) scales. Qualitative interviews focused on experiences living with CHD, social and family support, and executive functioning skills.

Results: Our mean participant age was 20 years, and 36 % had some college education. PROMIS scoring showed increased risk for poor physical function (mean $r = 31.8$), anxiety (mean $r = 64.9$), depression (mean $r = 66.8$), and pain interference (mean $r = 66.6$). AEFI results also showed poor executive functioning skills, particularly in Self-Control/Self-Monitoring (mean 18.1) and Planning and Initiative (mean 5.3). 27 % of participants had medical record documentation of depression. Interviews showed participants overinflated sense of coherence and poor executive functioning, leading to two main themes: 1) perceptions of their disability, including denial of their CHD diagnosis and disengagement with mental health needs, and 2) challenges in transition care.

Conclusions: Our study indicates a need for larger scale interventions for psychological young adults with CHD as part of the transition process to adult care.

1. Introduction

Congenital heart defects (CHD) are the most prevalent form of birth defect and impact between 4.0 and 9.3/1000 live births depending on severity of the defect and geographic location [1,2]. The birth prevalence of CHD has increased in the last five decades due to advances in earlier diagnostic technologies, but this varies regionally [1,3]. Improvements in medical and surgical treatment have decreased mortality in more severe CHD, allowing patients to live well into adulthood with appropriate monitoring and follow-up. These individuals are also

considered at high risk for developmental delay or disorders and may experience persistent delays leading to a diagnosis of developmental disability, cognitive dysfunction, or decreased intellectual capacity as measured by the intelligence quotient earlier in life [4,5]. The American Heart Association classifies individuals with cyanotic heart lesions, history of open heart surgery in infancy, a suspected genetic abnormality, and a list of other conditions and comorbidities as higher risk for these delays [6–9]. There is an increased risk for internalizing and externalizing behaviors in individuals with CHD compared to their typically developing peers aged 6 years and older [5].

* Corresponding author.

E-mail address: Smathena@arizona.edu (S.A. Mathena).

¹ These authors take responsibility for all aspects of the reliability and freedom from bias of the data presented and their discussed interpretation.

Less is known about these longer-term impacts on young adults with high-risk CHD who were not identified during youth, and findings on the extent of poor neurodevelopmental and psychological outcomes have been mixed. Patients with CHD tend to have an inflated sense of well-being and report lower psychological concerns as well as higher quality of life than their peers not living with a chronic illness [10]. This suggests some individuals with CHD, especially in adolescence, may mask psychological symptoms and go undiagnosed for conditions that lead to reductions in executive function skills and maladaptive coping mechanisms that further hinder medical compliance and ability for self-management of CHD. These unaddressed neuropsychiatric issues consequently put young adults at unnecessary increased risk for mortality and morbidity [11].

Despite a growing understanding of the neurodevelopmental delays in individuals with CHD, there is little understanding or consensus on the cognitive and psychological functioning of adolescents and adults with CHD despite risk of additional comorbidities as they age, particularly when left untreated [8]. Multiple authors have articulated the importance of more research on neurocognitive and psychosocial symptoms and its implications in adults with CHD, particularly for those with critical CHD's [10–15]. Wilson (2015) describes the need for clearly defined patterns of these manifestations in adults to maximize interventions and mitigate symptoms [16]. The Adult Congenital Heart Association's (ACHA) multidisciplinary working group list cognitive and psychiatric outcomes in adults as a critical area for research. Therefore, this study identifies and describes the psychosocial profile of a cohort of young adults (ages 18–24) diagnosed with CHD. Individuals were assessed via psychometric questionnaires, in-depth semi-structured interviews, and clinical assessments for their risk factors for executive dysfunction, maladaptive coping, and other psychological comorbidities.

2. Methods

2.1. Study Design

Participants were recruited either directly from the pediatric and adult congenital cardiology clinics or through outreach via social media and the ACHA website. Individuals were included if they were diagnosed with either moderate or severe CHD and/or at high risk for neurodevelopmental delays defined as having 1) open heart surgery as an infant; 2) a cyanotic CHD; 3) exposure to environmental risks in infancy (prematurity, resuscitation, mechanical support, prolonged hospitalization); or 4) traditional risk factors for developmental delay (genetic syndromes, abnormal CNS imaging, microcephaly, seizures). We restricted inclusion to ages 18–24 years as this age range constitutes legal adulthood but occurs prior to the predicted age of frontal lobe maturation when poor decision-making due to limited executive function skills is common [17]. Participants were asked to consent to all portions of the study, including a medical records review, a self-reported psychometric measures survey, and a semi-structured interview. This study was approved for human subjects research via the University of Arizona Institutional Review Board as protocol number #1804450559.

2.2. Measures

Psychometric measures included coping strategies (Responses to Stress Questionnaire), self-rated physical and emotional well-being (Patient-Reported Outcomes Measurement Information System), and executive function skills (Amsterdam Executive Function Inventory).

2.2.1. Responses to Stress Questionnaire: Pediatric Heart Problems (RSQ)

The RSQ was initially developed and validated in 2000 to measure patterns of coping and assess involuntary stress responses. Participants are provided a list of stressors related to their cardiac diagnosis and asked to record the frequency with which each stressor is experienced.

Participants are then asked to rate on a scale of 1–4 how often they utilize certain coping strategies when faced with those stressors. Scores comprise a proportion summing to 100 % for three types of voluntary coping and two types of involuntary coping [18]. Voluntary coping includes taking control through problem-solving and emotional modulation (primary); positive thinking and cognitive restructuring (secondary); and avoidance, denial, or wishful thinking (disengagement). Involuntary mechanisms include physiological or mental arousal (engagement) or mental numbing (disengagement). Involuntary mechanisms and voluntary disengagement can become maladaptive. The RSQ provides a scoring algorithm using SPSS [19].

2.2.2. Patient-Reported Outcomes Measurement Information System (PROMIS)

PROMIS is a validated tool measuring physical, mental, and social well-being. We are using the PROMIS-29 profile measures, which includes 4 questions within 7 domains (physical function, anxiety, depression, fatigue, sleep disturbance, social involvement, pain interference) plus an 11-point pain intensity scale [20]. Data was scored using the HealthMeasures Scoring Service, which returns a T-score normed by age and sex to the general population for each subscale. Typically, physical and mental health scores are not part of the standardized scoring package, however, accepted methods for generating these scores have been published [21].

2.2.3. The Amsterdam Executive Function Inventory (AEFI)

Validated by psychometric analysis, the AEFI assesses executive function components of everyday behavior including attention, planning and initiative, and self-control/self-monitoring. Results are normed to education level and gender. Participants are asked 13 questions answered on a 3-point Likert scale with scores reported as a percentage of the population at or below participant performance for the three domains [22].

2.3. Medical Record Abstraction

Medical records were used to confirm CHD diagnosis. Physician visit notes were transcribed from the medical records verbatim when referencing well-being or mental health by the first author and verified by the second author.

2.4. Semi Structured Interview

Following completion of the surveys, participants were interviewed using a HIPAA compliant video conferencing platform. The interviews were semi-structured with specific questions related to experiences living with CHD, social and family support, and executive functioning skills. Each interview lasted 45 min to 1 h and was transcribed using a naturalized transcription process for analysis. We used MAXQDA software [23] for qualitative analysis and coding using the King and Horrocks (2010) content analysis approach [24] with a particular focus on discrepancies between the patients' self-reported questionnaire data, their self-described lived experiences of CHD, including their social environment, and the statements regarding well-being and mental health as made by their medical provider.

2.5. Statistical Analysis

Descriptive analyses used means and standard deviations for continuous variables and counts and proportions for categorical variables. Correlation analysis was used to assess the linear relationships between the scale scores. Variables that were associated with the physical and mental health calculated summary scores were included in regression analyses. Data were analyzed using R [25].

3. Results

Demographics, CHD diagnoses, and distribution of participation in the study can be seen in Table 1. The mean age of participants in the study was 20 years old, and most had completed high school and at least some college education. All participants were from Arizona or the southwestern United States. 11 participants completed the surveys, 7 individuals had medical records available for review of physician notes on mental health concerns, neurodevelopmental delays, family medical histories, and confirmation of their CHD diagnosis, and 5 individuals participated in the qualitative interview. All participants self-reported as having either a moderate or severe CHD and an inclusion risk factor for participation in the study. Two surveys were sent anonymously and included only in the self-report measures.

Medical records and participant interviews were reviewed for self-reported family medical history of CHD's or neurodevelopmental delays. Two participants had unknown family history due to adoption as infants, but all other participants reported no relevant family history of CHD, mental health conditions, or neurodevelopmental delays. No participant had reported developmental delays, although 3 individuals did have medical record documentation of depression and 2 participants

Table 1
Demographic data on participants based on socioeconomic status including gender, age, occupation, marital status, and level of participation within this study.

	N	%
Total Participants	11	100
CHD Diagnosis		
Moderate		
Bicuspid aortic valve with aortic regurgitation	1	
Bicuspid aortic valve with aortic stenosis	1	
Coarctation of the aorta and bicuspid aortic valve	1	
Coarctation of aorta and Ventricular Septal Defect	2	
Ventricular and Atrial Septal defects	1	
Specific Diagnosis unknown	2	
Complex		
Single left ventricle, pulmonary atresia	1	
Tricuspid Atresia with Fontan	1	
Taussig-Bing syndrome	1	
Race-Ethnicity		
White	1	9
Hispanic or Latino	2	18
American Indian or Alaskan Native	1	9
Other	1	9
Not Answered	6	55
Work and/or School		
Work	4	36
School	6	55
Not Answered	4	36
Gender		
Male	4	36
Female	3	27
Not Answered	4	36
Education		
Completed High School	2	18
Some College	4	36
Completed College	1	9
Not Answered	4	36
Marital Status		
Single	5	45
Not Answered	6	55
Insurance Status		
Private	5	45
Medicaid	3	27
Unknown	3	27
Medical Record Reviewed	7	64
Participated in Interview	5	45

	Mean	Range
Age	20	(18–22)

had records of suicidal ideations. Insurance status was reviewed as a proxy for income status during medical records review and the interview, and 45 % of the participants had private insurance and 27 % reported coverage under Medicaid, and 27 % were unknown.

Table 2 shows the mean and standard deviation of the psychometric survey data.

Overall, the participants' mean scores were at increased risk for poor physical function, anxiety, depression, and pain interference. Results also showed poor executive functioning skills, particularly in Self-Control/Self-Monitoring and Planning and Initiative. The correlation matrix in Table 3 demonstrates a negative association between anxiety and depression and physical functioning and a positive association with fatigue ($P < 0.05$). Pain was positively associated with disengagement coping, whereas physical functioning was negatively associated. Disengagement coping is negatively associated with overall physical and mental health in both correlation and regression analyses (Fig. 1).

Analysis of the interviews identified two key themes: 1) Perceptions of disability, which has two subthemes of disengagement with their mental health needs and denial of the CHD as a chronic concern; and 2) Challenges in transition, which revolved around executive dysfunction in management of their health care.

4. Discussion

Mood and anxiety disorders are triggered by various biopsychosocial factors, [26] including genetics, adverse life experiences, and cognitive dissonance. Individuals with CHD have increased risk for anxiety and mood disorders, and current estimates are close to 1 in 5 adults carrying a diagnosis of depression [27]. Mental health issues have been understudied in young adults with CHD. Our data show that many of the participants in our study struggled with maladaptive coping when dealing with issues associated with their CHD more than half of the time ($M\ 55.1$, $sd\ 10.3$) when faced with a stressor. Maladaptive coping includes active disengagement strategies and involuntary engagement and disengagement, and our participants' perceptions of disability contributed to these disengagement coping behaviors, including efforts to avoidance of the topic, denial of illness or disability, and wishful thinking [28,29].

Other than the potential thoracotomy or sternotomy scars, CHD is an invisible disease, and many youth in our study elected to ignore or avoid

Table 2
Means and standard deviations of coping and symptom variables from each survey instrument within the study.

	M	SD
Response to Stress Questionnaire		
Perceived Stress Score from having a cardiac condition	25.1	11.7
Primary Control Coping Ratio (i.e., problem solving, emotional expression, emotional modulation)	16.0	4.6
Secondary Control Coping Ratio (i.e., positive thinking, cognitive restructuring, acceptance, distraction)	28.2	6.8
Disengagement Coping Ratio (i.e., avoidance, denial, wishful thinking)	17.4	1.8
Involuntary Engagement Coping Ratio (e.g., physiological arousal, rumination)	21.1	5.1
Involuntary Disengagement Coping Ratio (e.g., emotional numbing)	16.6	3.4
Amsterdam Executive Function Index		
Attention	60.6	21.2
Self-Control and Self-Monitoring	18.1	22.6
Planning and Initiative	5.30	8.59
PROMIS Scale T Scores		
Physical Function	31.8	13.4
Anxiety	64.9	11.4
Depression	66.8	15.4
Fatigue	55.1	12.8
Sleep Disturbance	54.0	5.5
Pain Interference	66.6	12.1
Ability to Participate in Social Roles and Activities	56.0	10.7

Table 3

Correlation matrix of the survey results and symptoms domains from each survey instrument.

	PROMIS							RSQ						AEFI		
	P H Y	A N X	D E P	F T G	S L P	S O C	P N	S T R	P C C	S C C	D C	I E C	I D C	A T T	S C S M	P I
Physical (PHY)	1.00															
Anxiety (ANX)	0.65	1.00														
Depression (DEP)	0.87	0.80	1.00													
Fatigue (FTG)	0.55	0.69	0.63	1.00												
Sleep (SLP)	0.10	0.19	0.00	0.18	1.00											
Social (SOC)	0.06	0.40	0.15	0.20	0.12	1.00										
Pain (PN)	0.98	0.58	0.82	0.55	0.01	0.05	1.00									
Stress (STR)	0.03	0.27	0.18	0.07	0.09	0.84	0.10	1.00								
Primary Control Coping (PCC)	0.19	0.52	0.17	0.44	0.62	0.44	0.21	0.28	1.00							
Secondary Control Coping (SCC)	0.18	0.57	0.22	0.11	0.07	0.53	0.05	0.53	0.22	1.00						
Disengagement coping (DC)	0.73	0.58	0.55	0.66	0.03	0.51	0.69	0.25	0.34	0.38	1.00					
Involuntary Engagement coping (IEC)	0.17	0.28	0.13	0.10	0.43	0.46	0.23	0.52	0.74	0.48	0.08	1.00				
Involuntary Disengagement coping (IDC)	0.13	0.66	0.27	0.32	0.53	0.81	0.07	0.69	0.72	0.69	0.45	0.68	1.00			
Attention	0.27	0.08	0.00	0.46	0.42	0.02	0.19	0.24	0.27	0.48	0.16	0.10	0.07	1.00		
SCSM	0.19	0.67	0.51	0.64	0.15	0.54	0.09	0.50	0.22	0.59	0.42	0.14	0.61	0.22	1.00	
PI	0.04	0.18	0.08	0.07	0.05	0.89	0.15	0.81	0.37	0.59	0.51	0.47	0.70	0.25	0.46	1.00
PHSum								0.10	0.23	0.22	0.77	0.13	0.21	0.26	0.24	0.04
MHSum								0.29	0.55	0.36	0.76	0.11	0.62	0.26	0.72	0.29

P<0.05

P<0.01

P<.001

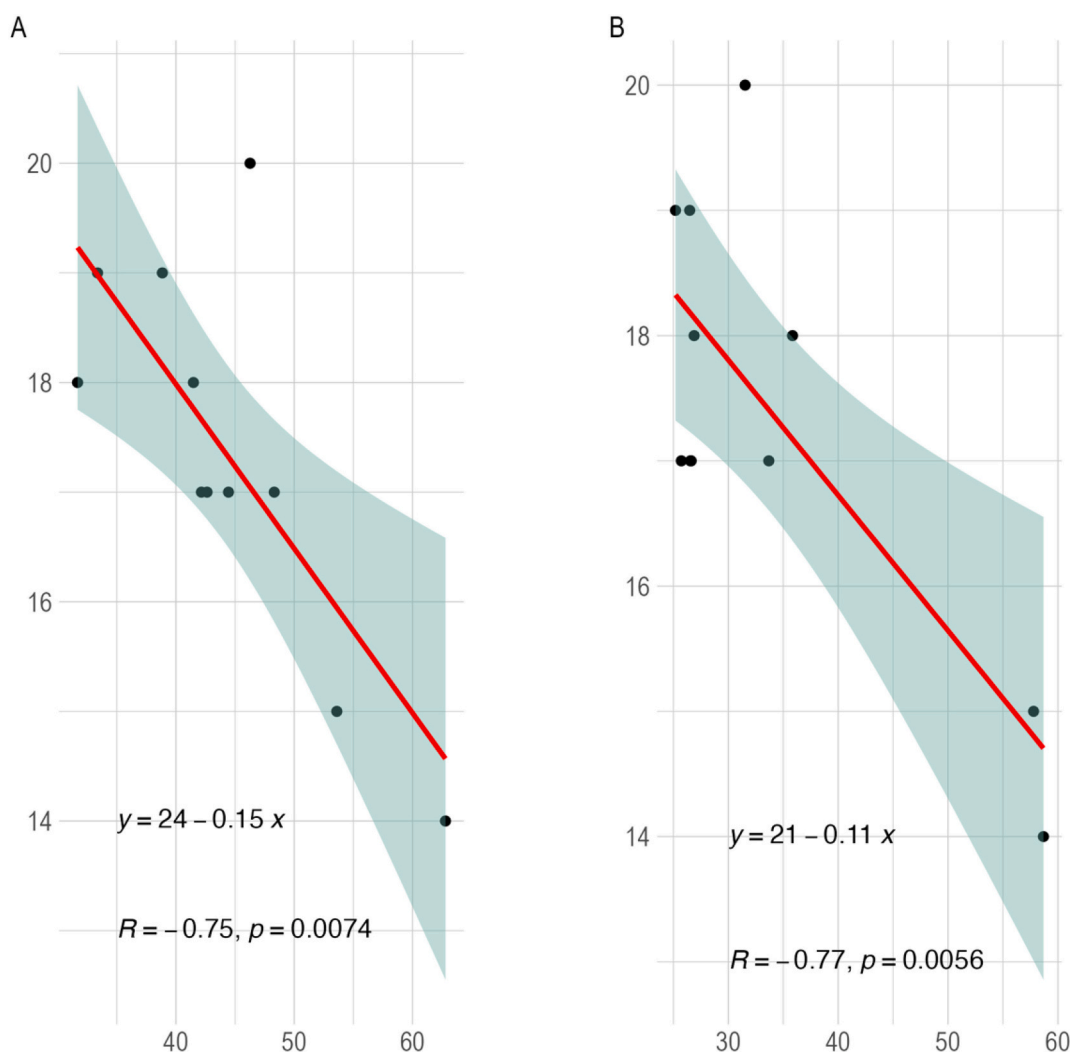


Fig. 1. Disengagement coping is negatively associated with both overall mental and physical health based on survey responses.

any issues related to poor health or the need to behave differently than their peers. Our participants frequently struggled with avoidance both in their daily lives and with their management of their CHD ($M = 17.4$, $sd = 1.8$), which was reflected in their interview responses to questions regarding their experiences living with CHD. Participants characterized their physical state as “fine” or “normal” or “just like everybody else” despite having a CHD, with one participant failing at a job because they declined to request accommodations from their employer so as not to be perceived as physically incapable. When asked, ‘how well do you think you manage your CHD?’ most patients stated that they managed their condition extremely well despite reporting they rarely thought about or did not want to think about their CHD reporting “just ignoring things until they go away” or “trying not to think about things.”

Wishful thinking resulted in potentially problematic behavior. One participant described going on a 7.5 mile hike with friends and feeling frustrated by their concern about her CHD, despite never previously having hiked longer than 3.5 miles.

So, like they were able to see that I would let them know if I needed help if I wasn't doing well, I knew my limits... But I also believe that it's ok to push your body sometimes and to push yourself mentally to go farther than you think you can... Yeah, so normally, I never try to call it quits.

Participant 2

Another participant decided to increase their physical activity after a nurse stated concerns.

I had a stress test, and the nurse was saying I am going to get a stroke. So, I started going to the gym after that because I didn't like my results even though they were like good, or they were normal for people like me.

Participant 1

Other aspects of denial include frustrations with having to “do all this other stuff” that their peers are not required to do and discounting the concerns of others because they believe what is normal for them should always be viewed as normal despite clinician concerns.

I have had it [CHD] since I was born and so I feel fine. And I also feel weird that I am not supposed to be fine. I have a very low blood pressure. Kinda abnormally low. So, they always ask, “Are you feeling, ok?” And I say, “Yeah, why?” I feel fine?”

Participant 4

Participants' scores showed struggles with other forms of maladaptive coping mechanisms, including a high proportion of involuntary engagement coping ($M = 21.1$, $sd = 5.1$), which includes rumination or physiological arousal response. Involuntary disengagement coping, such as emotional numbing, was also high in our sample ($M = 16.6$, $sd = 3.4$). Both of these maladaptive coping responses are commonly present in individuals with anxiety and depression, which may either be caused by or contributing to mental health conditions such as anxiety and depression [28]. When questioned during interviews regarding their feelings about living with a CHD, participants frequently expressed

feelings of numbness or even confusion,

“I feel like I have a broken heart. I don’t know if I am supposed to feel strong or weak with it.”

Participant 4

Four out of five participants reported no concerns with anxiety or depression, yet the majority demonstrated scores above the cutoff for concern for both anxiety and depression on the scales. Research has shown that there is increased risk for mood and anxiety disorders in ACHD, but underdiagnosis and undertreatment persist [13]. Three of our individuals had a documented diagnosis of depression on medical records, including suicidal ideation in two participants. One participant reported “worrying all the time” but described themselves as “mentally strong.” Another indicated that they did not experience depressive symptoms yet reported “sleeping the vast majority of [each] day.” Participants experienced feelings of isolation and loneliness. Many of these concerns also came from the limited social support experienced by participants, either from family or friends currently or in the past. Despite reporting high levels of parental involvement and support, many reported a nonexistent social life outside of their family unit or only having a limited number of friends, or “preferring not to deal with people.” Family and social circumstances such as divorced parents or unstable housing also frequently alienated participants from means of peer social support. Participants reported issues with being ‘overly shy’ or ‘hard to open up to people.’ Only one participant reported feeling like her current social group at church was beneficial for her current mental health despite having some social struggles in the past.

Another potential contributor to both maladaptive coping and depression is the high level of potential executive dysfunction we saw in our participants. Our participant scored at or below 18 % of their peers in self-control and self-monitoring and 5 % for planning and initiative. When asked, many participants frequently struggled with elements of executive function, specifically in memory and task avoidance. Participants frequently described traits that exhibited poor executive functioning skills, like refusing to talk to parents regarding life goals and having no clear future planning or life goals such as college or future careers. During the interview process, participants were asked questions regarding how well they manage their to-do lists and tasks associated with the care of their CHD, particularly medication adherence. The majority of the patients self-reported either during the interview or to their clinician that they struggled with self-managing their medications for CHD. Patients reported during the interview that medication compliance was challenging as they do not see the need to take medications when they “feel completely fine,” or “don’t feel any different without it.” Parental reminders to take medications were frequently met with annoyance or dismissed as being “overly paranoid,” as stated by one participant.

Our participants struggled to define their form of CHD, even in layman’s terms, and could not articulate their management needs. Additionally, most participants completely lacked the social and clinical support to fully take over their clinical care without a parent or guardian present. One participant even reported not understanding she had a CHD until she turned 18 and needed additional surgery. Participants did report positive interactions with their cardiologists and appreciated their provider “talking to me like an adult” about managing their care. However, many of these discussions were reported after the participant turned 18 and had begun the transition into college. Only one participant had reported attending a doctor’s appointment without a parent. Many of our participants wanted to achieve goals such as college degrees but lacked a plan to do so successfully. Participants reported statements like “preferring just to stay home” or wanting to finish college to please a parent but “not being motivated to go” even when desiring to have a college degree to feel successful in their life.

Transition of young adults with chronic conditions has become the focus of many medical providers and agencies, but outcomes of transition programs are often inconsistent due to timing of transition,

coordination of care, and the role of parental involvement [30,31]. Similar concerns in other young adults with chronic conditions have also been reported in CHD patients, including lack of patient knowledge of their condition, poor adherence to medical advice, and overall poorer health outcomes [32–34]. Gaps in care when transitioning out of pediatric care have been commonly recognized amongst ACHD populations, particularly in those patients who report ‘feeling well’ or were unaware that they would need additional medical care [35], leading to poorer health outcomes long term [36].

4.1. Study Limitation

Our study is a single center study of a small cohort study of young adults that may not be generalizable to larger populations. Study data was obtained via retrospective chart reviews that may not be able to determine causality/etiology of psychiatric disorders. Surveys were collected via mailed and electronic survey formats, and interviewers were collected via teleconference interview software and may be subjected to recall bias or unwillingness to disclose by the participants.

5. Conclusions

Results from our study show high rates of mental health comorbidities amongst our participants. Longitudinal analysis of the results from different measures of this study illustrates that a majority of participants had unmet mental health needs that were increasing as they aged and lived. Survey results suggest participants frequently engaged in disengagement coping strategies such as avoidance or denial of challenging social and personal situations, such as social relationships. Similar results were found in the psychometric measures regarding when participants were asked questions regarding their CHD management and care.

Despite reporting doing well with management and understanding of their CHD, participants in this study self-reported frequent mismanagement of their health care, such as missed medications or failure to comply with medical advice. Results from qualitative analysis of the interviews suggest that young adults with CHD may struggle to manage their condition but also are hesitant to be labeled or identified as a young adult with a health problem. Particularly, our findings suggest that young adults with CHD may be hesitant to follow medical advice as they do not want to be seen as different than their peers. Research has suggested that maintaining care is critical to optimize health outcomes for young adults with CHD, as loss to follow-up care increases the risk of additional disease morbidity and mortality, including additional hospitalizations [11].

Our study suggests that a better and earlier emphasis needs to be placed on transition-based care, both for their CHD management and mental health comorbidities. Currently, the AHA recommends transition care begin at approximately age 12. However, recent studies on transition readiness in young adults have shown deficits in the essential medical and lifestyle knowledge for CHD management and care [32,37]. However, patients with non-severe CHD exhibit little to no risk factors within the clinical settings for neurodevelopmental disorders and thus may be missed in this transition process. Many of the risk factors for mental health comorbidities amongst this population emerged likely due to maladaptive coping mechanisms that had long-term impacts on their physical and mental well-being. Research has shown that improving self-efficacy and self-determination in transition care improves the quality-of-life outcomes in chronic conditions. However, many young adults with complex mental needs or with emotional, behavioral, or developmental disabilities are much less likely to receive adequate transition care [38].

We propose self-efficacy and self-determination must be included as part of the transition process for individuals with CHD to improve self-management of their condition and help overcome these maladaptive strategies. Our study illustrates that the current transition process for many young adults with CHD is insufficient for empowering young

adults with CHD to manage their own conditions into adulthood. Consequently, more needs to be done in the transition process for young adults with CHD to improve their medical compliance and decrease mortality risk as they transition into adult-based care. Specifically, we suggest directed health management education, on the need for lifelong management of CHD, physician coaching to encourage self-efficacy and medical compliance as they transition to adult care, particularly for patients at risk of not meeting transition outcomes. This process should also include periodic screenings and monitoring of neurodevelopmental and psychosocial, as patients move into the transition process as executive functioning and mental health concerns can increase as the patient ages. Parental guidance on how to gradually release medical responsibilities as their child transitions to improve patient self-determination and efficacy in managing their own medical care into adulthood. Finally, training for pediatric cardiologists on screening and services available for patients in need of behavioral and mental health services [39].

The relationship between CHD and poor psychosocial and neurological outcomes is dependent upon several risk factors such as genetics, parental stress, socioeconomic status, access to consistent medical care, and other factors. Future research foci include assessing these risk factors across different demographics of CHD populations and identifying tools for intervention for populations at most risk for poorer neurodevelopmental and mental health outcomes.

CRedit authorship contribution statement

Sarah A. Mathena: Writing – review & editing, Writing – original draft, Validation, Project administration, Formal analysis. **Vanessa F. Eller:** Writing – review & editing, Investigation. **Scott E. Klewer:** Writing – review & editing, Supervision, Resources. **Jennifer G. Andrews:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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